RecallX Pro: System Architecture Report

1. Executive Summary

RecallX Pro is a highly engaging, dynamically scaling sequence memory game designed for modern web browsers. It challenges players' cognitive retention by generating increasingly complex "Simon Says" style memory patterns across a responsive geometric grid. The application is built from the ground up to be a completely serverless, production-grade enterprise web application, featuring secure email authentication and global competitive tracking.

2. Core Feature Specifications

2.1 Dynamic Gameplay Mechanics

- **Algorithmic Scaling:** The game consists of 5 distinct levels. The system dynamically expands both the grid size (from 8 tiles to 36 tiles) and the sequence length based on the player's progression.
- **Fail-Safe Forgiveness:** Unlike traditional strict memory games, players are not immediately eliminated upon a wrong click. Instead, the game tracks `failures`, dynamically alerts the user through CSS vibration animations and red color warnings, and allows them to continuously retry the sequence.
- **Progress Tracking:** A real-time progress bar monitors sequence completion cleanly communicating exact progression milestones.

2.2 Secure OTP Verification

- **Email Authentication:** The application features a password-less login mechanism. Players enter their Name and Email address, triggering a server-side action that generates a random 4-digit One Time Password (OTP).

- **Automated SMTP Dispatch:** The Vercel cloud architecture natively builds an HTML-formatted secure email containing the code and dispatches it directly to the user's personal inbox via a secure NodeMailer dependency.

2.3 Global Cloud Leaderboard

- **Competitive Sorting:** The system evaluates players primarily on **Highest Level Achieved**, utilizing **Fewest Failures** as a mathematical tie-breaker algorithm.
- **Privacy Truncation:** To adhere to modern data privacy standards, the Leaderboard interface intelligently sanitizes email addresses (e.g., `ved***@gmail.com`) before broadcasting scores to the public dashboard.

3. Technology Stack & Deployment

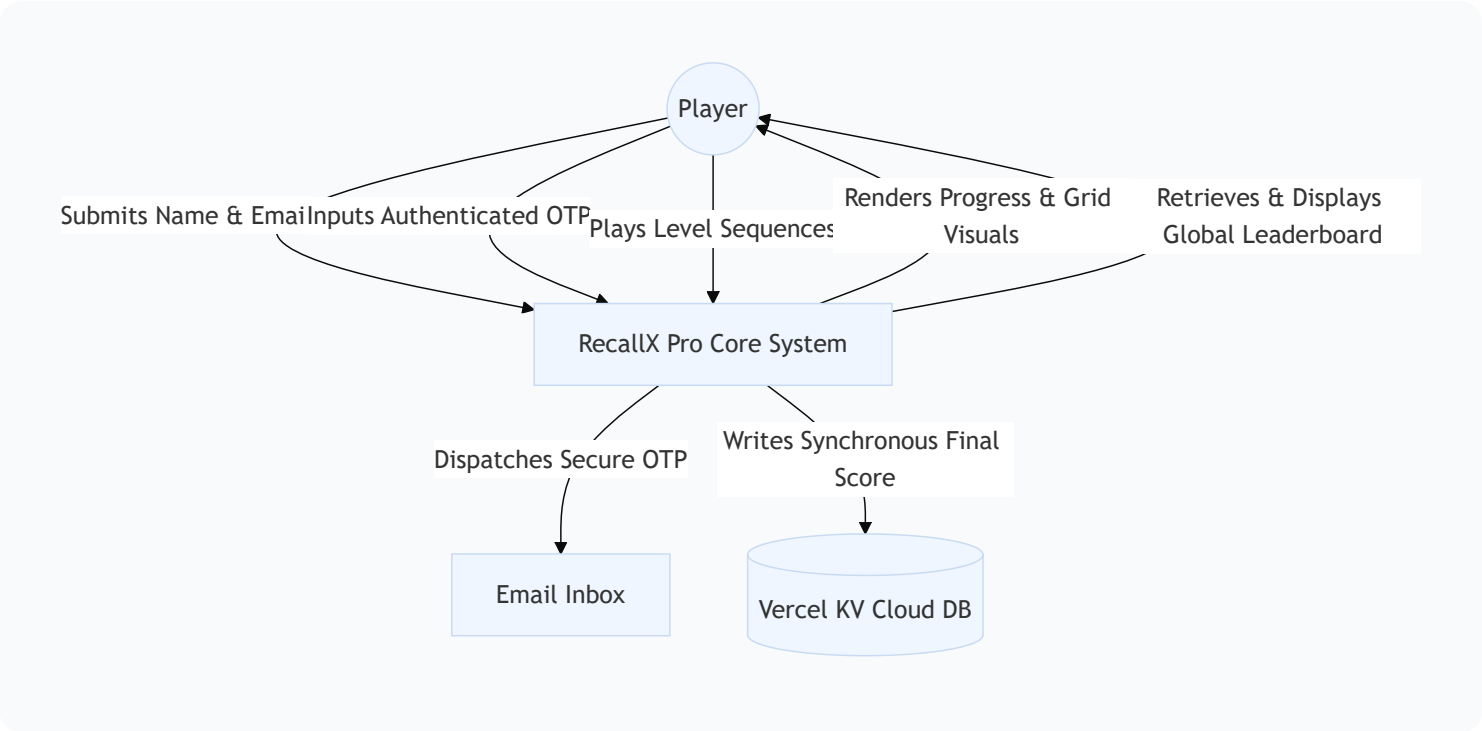
Logical Layer	Technologies Used	Description
Frontend Interface	HTML5, CSS3, ES6 JS	Designed with advanced Glassmorphism CSS architecture, animated `@keyframes float` background blobs, and deeply responsive CSS Grids.
API Endpoints	Node.js (Serverless)	Executes ephemeral computing tasks cleanly bypassing physical server limitations.
Messaging API	NodeMailer	Replaces standard costly SMS integrations with completely free SMTP email routing.
Cloud Database	Vercel KV (Redis)	Stores the massive Global Leaderboard utilizing strict JSON string arrays and ultra high-speed memory reads.
Pipeline Deployment	Vercel & GitHub Actions	Complete CI/CD implementation ensuring code updates pushed vertically dynamically patch the live internet environment effortlessly in seconds.

4. Data Flow Diagrams (DFD)

The following diagrams illustrate how information fundamentally moves through the RecallX hardware and network ecosystem.

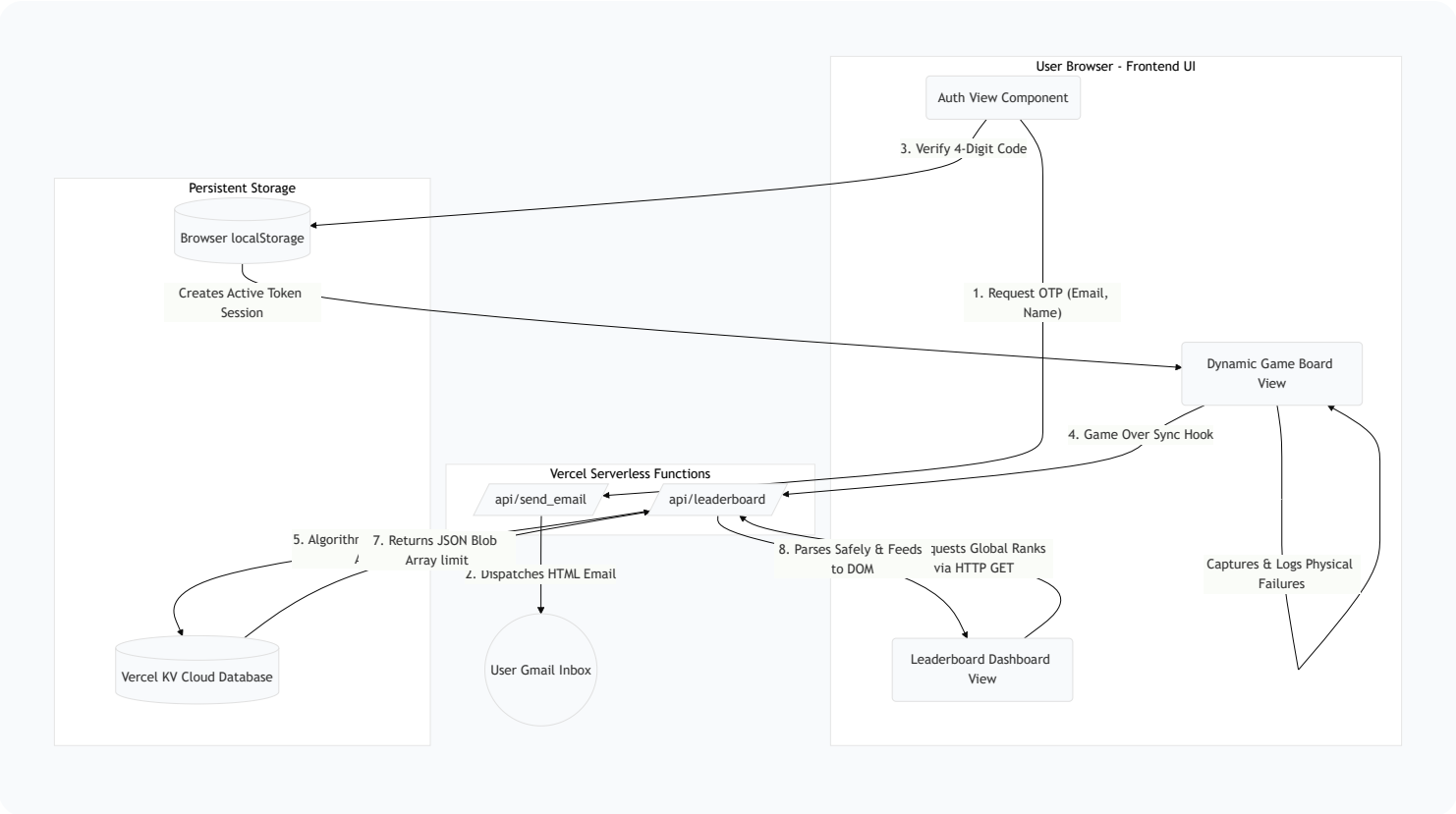
Level 0 Context Diagram

This diagram highlights the highest-level interaction between the External Human Player and the Core Application System.



Level 1 DFD: Game Execution & Network Flow

This diagram details the internal server processes and how the application parses memory, HTTP routing, and API connections.



5. Enterprise Security & Limitations

- **Serverless Cloud Secrets:** All dangerously sensitive variables (specifically Google SMTP Application Passwords and complex Cloud Database Cluster Keys) are strictly contained securely inside Vercel's Cloud Environment Variables. The public interface execution layer (`script.js`) physically cannot expose these credentials keeping all systems fully resilient.
- **Network Fail-Safe Fallbacks:** If the environment ever drops connectivity to the underlying Serverless API paths due to a catastrophic outage, the logic explicitly hooks into integrated `try..catch` safety nets. Rather than hard-crashing and locking users out, the system triggers graceful degraded visual fallbacks simulating the inbox components visibly on-screen to ensure continuous offline app mobility.

Prepared for Client Presentation — Compiled dynamically from RecallX Systems.